In-Flight Spectral Response Function Retrieval of a Multispectral Radiometer Based on the Functional Data Analysis Technique

Na Xu, Peng Wu, Gang Ma, Qirui Hu, Xiuqing Hu, Ronghua Wu, Yunfeng Wang, Hanlie Xu, Lin Chen, and Peng Zhang

***Abstract*— The spectral response function (SRF) is a crucial parameter in multispectral radiometers, and it influences the radiometric calibration accuracy and quantitative application capabilities. The in-flight SRF often has errors due to prelaunch contamination or postlaunch degradation. This study proposes an innovative new method to retrieve SRFs of multispectral radiometers based on intercomparisons with hyperspectral sounders via the functional data analysis (FDA) technique. Under the FDA framework, all variables, including the hyperspectral radiance and SRF, are regarded as functions rather than discrete data by expanding in the Fourier functional basis. The forward convolution equation is processed directly into a functional integration model rather than a normally pointwise summation; this ensures that the unknown quantities are transformed from numerous SRF samples to several function parameters, thus avoiding the ill-posed problem. The proposed algorithm is verified with both simulated and real data from multiple thermal infrared bands of the FY-3 IRAS and FY-4 AGRI using collocations with METOP-B IASI. All these results demonstrate our algorithm's qualitative and quantitative effectiveness for infrared SRF retrieval. Although the demonstrations are particularly relevant to infrared spectra, the algorithm is universal and also applicable to other spectral bands.**

***Index Terms*— Functional data analysis (FDA), intercalibration, multispectral radiometers, retrieval, spectral calibration, spectral response function (SRF).**

I. Introduction

THE spectral response function (SRF) is a vital parameter of optical remote sensors, and it comprises the comprehensive performance of the reflection and transmission characteristics of each optical component and the spectral response characteristics of the detectors. The SRF determines the characteristics of the spectral response of the remote sensor to radiation, which directly affects the observed target radiation and the reflected or emitted radiation from the onboard calibrator. Quantitative applications of remote sensing observations require the use of spectral responses to simulate the top of atmospheric radiation of the corresponding sensor band. With an increasing demand for quantitative remote sensing and improvements to instrument performance, increasingly stringent requirements are needed for the overall calibration accuracy. At present, a considerable amount of progress has been made in radiometric calibration, thereby causing the accuracy of spectral calibration to become increasingly important. This is especially the case for multispectral optical sensors, where both the spectral shape and center position of the SRF have an important effect on the spectral calibration accuracy. As most multispectral radiometers do not have an onboard spectral calibration device, the spectral response is typically characterized by a prelaunch laboratory spectral calibration and is kept constant over its lifetime. Numerous studies have revealed that postlaunch spectral response characteristics are also likely to change because of certain factors such as the space environment and temperature variations of the satellite instrument. For example, spectra shifting has been observed in the thermal infrared absorption bands of MODIS [1]–[3], HIRS [4], [5], and the GOES series satellites [6]–[10], with shifts ranging from a few to more than 10 wavenumbers. Because of water–ice contamination, the instrumental spectral characteristics of the GOES and METEOSAT geostationary satellites have been altered, resulting in continuous variations during all decontamination procedures [11], [12]. Furthermore, certain spectral errors originate from prelaunch measurements. When limited by measurement facilities, the laboratory spectral response measurement may be contaminated by atmospheric absorption between the spectral calibration system and target radiometer. This contamination can cause the SRFs of the water vapor (WV) bands on the Japanese and Chinese Geostationary Meteorological Satellites (GMS-5 and FY-2, respectively) to exhibit similar distributions with laboratory atmospheric transmission spectra, which leads to an error of more than 1k for the simulated radiance [13]–[15].

Manuscript received March 11, 2020; revised June 23, 2020, September 25, 2020, and December 7, 2020; accepted January 25, 2021. Date of publication April 28, 2021; date of current version December 20, 2021. This work was supported in part by the National Key Research and Development Program of China under Grant 2018YFB0504800, Grant 2018YFB0504802, and Grant 2018YFB0504900; and in part by the National Natural Science Foundation of China under Grant 41471304. *(Corresponding author: Na Xu.)*

Na Xu, Gang Ma, Xiuqing Hu, Ronghua Wu, Hanlie Xu, Lin Chen, and Peng Zhang are with the Key Laboratory of Radiometric Calibration and Validation for Environmental Satellites (LRCVES/CMA), National Satellite Meteorological Center, China Meteorological Administration, Beijing 100081, China (e-mail: xuna@cma.gov.cn).

Peng Wu is with the Beijing International Center for Mathematical Research, Peking University, Beijing 100871, China.

Qirui Hu is with the Center for Statistical Science, Tsinghua University, Beijing 100084, China, and also with the Department of Industrial Engineering, Tsinghua University, Beijing 100084, China.

Yunfeng Wang is with Hohai University, Nanjing 210098, China.

Digital Object Identifier 10.1109/TGRS.2021.3073097

1558-0644 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:21:19 UTC from IEEE Xplore. Restrictions apply.

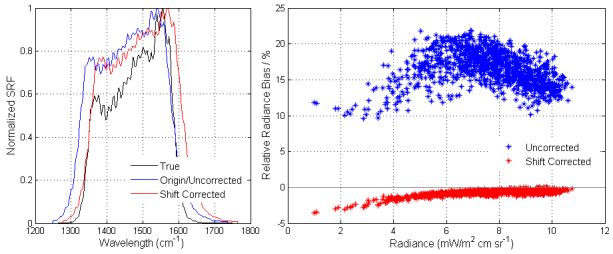


Fig. 1. Schematic of a simulation experiment of the effect of the SRF error and shift correction based on hyperspectral observation data. (Left) SRFs comparison. (Right) Relative radiance bias distribution.

To date, previous studies have developed several approaches to solve the error associated with the SRF. For contamination issues, if the root source, such as water-ice or laboratory WV, and its optical properties are known, the spectral error can be directly corrected by removing contamination absorption using the transmittance spectra ratio [11], [13], [15]. For other more general problems, the most adopted method is spectral shifting, which is a practical simplification to model complex SRF changes. Once the error is identified, the SRF is shifted until there is either a minimum absolute radiance bias or a minimum absolute scene-dependent slope occurs with respect to the reference observation [5], [10], [16]. The spectral shifting method attributes spectral errors to the frequency drift, and ignores the effects of spectral distortion which may also be important for the broadband radiometer. Fig. 1 shows the result of a simulation experiment where the true SRF is artificially deformed and shifted (regarded as the origin/uncorrected SRF). For this case, the spectral error cannot be eliminated by shifting alone, and the residual bias has obvious scene-dependent characteristics.

Aiming to achieve SRF shapes correction, several studies have resorted to in-flight SRF retrievals. Sun *et al.* (2008) retrieved infrared SRF of AQUA MODIS using collocated hyperspectral observation data based on a constrained regression mathematical algorithm. The preliminary results showed that the constrained retrieval algorithm can, to a certain extent, overcome the noise sensitivity effect and provide an in-flight SRF. Nevertheless, as mentioned in [17], the involvement of the constraint leads to a bias from the "real" result, resulting in a compromise between an increased noise constraint and less constraint bias. Quast *et al.* [18] recently proposed a metrologically sound method for retrieving the VIS spectral response from matchups of pseudoinvariant calibration site (PICS) pixels with data sets of simulated top-of-atmosphere spectral radiance used as a reference. To characterize SRF aging, the absolute instrument SRF was decomposed into generalized Bernstein basis polynomials and a degradation function.

This study proposes a new method to retrieve the SRF by approaching it as a function rather than discrete data via a functional data analysis (FDA) technique. Recently, FDA has attracted considerable attention in various scientific communities, and it has become a prevailing method used in economics, biology, meteorology, psychology, and industry, among other subjects. FDA is a multivariate statistical analysis method that uses observed data as an element of an infinite dimensional functional space [19]–[22]. With the advance of modern technology, there is a continuously increasing amount of data recorded during a certain interval or intermittently at several discrete sample points, which are known as functional data. FDA views these discrete observed data comprehensively instead of solely from the order of individual observations to explore the intrinsic functional structure of the data; i.e., not its external representation. For remote sensing, FDA is generally used to investigate the variation characteristics of certain meteorological or environmental parameters at different periodic scales, which are characterized by a smooth continuous function over time [23]–[25]. For our case, if the SRF is featured as discrete samples, the retrieval program requires multiple formulas to resolve hundreds of unknown variables, i.e., finer sample intervals yield more variables, therefore inducing the ill-posed problem. The FDA method considers the SRF as an entire smooth function, describing it through basis functions and functional parameters. Thus, the variable is transformed from numerous SRF samples to several function parameters, which can substantially reduce the dimension of the unknowns and ensure a more stable and smooth solution.

This study introduces a new in-flight retrieval method for the multispectral radiometer SRF based on the FDA technique. The methodology of the proposed algorithm is described in Section II. Section III validates the experimental results. Parameter selection is discussed in Section IV. Finally, we provide our conclusion in Section V.

II. Methodology

A. Basic Formulas

The observation of multispectral radiometer can be simulated by convolving the hyperspectral observations with their corresponding SRFs, which can be expressed as follows:

$$L^y = \int L^x(\upsilon) \cdot \varphi(\upsilon) d\upsilon + \varepsilon \tag{1}$$

where L^y is the multispectral radiance, L^x is the hyperspectral radiance over the same target, ϕ is the integral normalized SRF (i.e., $\int \varphi(\upsilon) d\upsilon = 1$), υ is the wavenumber, and ε is the error term, including instrument noise, convolution error, and data collection error. In general, the convolution and data collection errors are small [26], [27], causing ε to be dominated by instrument noise.

Here we need to estimate $\varphi(\upsilon)$ based on the above model. Based on the limitations associated with data collection process, we are unable to continuously observe the value of $L^x(\upsilon)$; instead, we are only able to obtain the values in discrete spectral channels, i.e., $L^x(\upsilon_1), \ldots, L^x(\upsilon_s)$ denoted by $L^x_1, \ldots L^x_s$. We assume that the observed data are $\{L^y_i, L^x_{i1}, \ldots, L^x_{is}\}$ for $i = 1, \ldots, p$. Here s and p represent the number of discrete spectra and observation pixels, respectively.

B. Retrieval Models

1) Why FDA: Normally, continuous integration can be expressed as follows:

$$L^y_{p\times 1} = L^x_{p\times s} \cdot \varphi_{s\times 1} + \varepsilon_{p\times 1} \tag{2}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:21:19 UTC from IEEE Xplore. Restrictions apply.

where $L^y_{p\times 1}$, $L^x_{p\times s}$, $\varphi_{s\times 1}$, and $\varepsilon_{p\times 1}$ represent the multispectral observation vector, hyperspectral observation matrix, SRF vector, and error vector, respectively. We assume ε is the instrument noise and satisfies $E(\varepsilon) = 0$ and $\text{var}(\varepsilon) < \infty$. Using the least squares method, we can solve the following:

$$\hat{\varphi}_{s\times 1} = \underset{\varphi}{\text{argmin}} \left\| L^y_{p\times 1} - L^x_{p\times s}\varphi \right\|^2 \tag{3}$$

where the symbol || || is the operator for calculating the Euclidean distance (ED), and the same thereafter.

A major challenge with respect to the least squares solution is the inverse problem. Owing to high autocorrelation and collinearity in the sample dimension, L_x is rank deficient; inverting (3) is an ill-posed problem, which leads to unstable regression results that are highly sensitive to noise. To alleviate this problem, a commonly used technique is regularization or shrinkage by adding a penalty or regular term to regularize the parameters. The penalty function can generally be expanded as follows:

$$\hat{\varphi}_{s\times 1} = \underset{\varphi}{\text{argmin}} \left\| L^y_{p\times 1} - L^x_{p\times s}\varphi \right\|^2 + \gamma \cdot J(\varphi) \tag{4}$$

where $J(\varphi)$ is the penalty or regular term and γ is a tuning factor that sets the weight of the penalty constraint. When $\gamma = 0$, the penalty term has no effect. As γ increases, the impact of the shrinkage penalty grows, and the estimate of $\hat{\varphi}_{s\times 1}$ will gradually depend on $J(\phi)$. Therefore, selecting an acceptable value for γ is critical. The penalty term can be a smoothing constraint, background constraint, or based on other methods. Generally, for atmospheric parameter retrieval, the background constraint scheme can be adopted, because the background has predictive indications for the future retrieval results [28]. However, the penalty function has limitations in SRF retrieval. First, if we use *a priori* background penalty, the inversion result will be highly relative to the choice of the penalty. Second, the smoothing constraint considers the local correlation at each point while ignoring the overall structure in $\varphi(\upsilon)$. Third, the penalty term will lead to bias from the "real" result. Moreover, the selection of γ and penalty also poses a problem in that it is artificial and can only be determined via empirical trials.

Physically, the SRF should be a nonnegative and smooth function. The discretization process will increase the uncertainty of the solution because of the limited characterization for the true variation, discrete sampling errors, and numerical solution instability caused by numerous variables. The FDA method, directly from integration model (1), regards the entire measurement as a smooth function, exploiting the FDA principle to solve the model. This method has several merits. On the one side, FDA considers the intrinsic correlation of discrete variables, and in turn makes the retrieval results obey a certain relationship that cannot be considered by the discretization process, which treats each discrete sample as an independent element. This can strongly benefit the retrieval by accommodating measurement noise and ensuring a smooth solution. On the other side, compared with penalty constraints, FDA does not require an initial value to a driven iterative solution, nor does it require *a priori* background or other artificial penalty constraints, which makes the algorithm more robust and has a wider practical application potential.

2) FDA Model: Equation (1) is a typical functional linear model that has been studied extensively [29]–[31]. The first step of the FDA is data preprocessing. Specifically, considering an orthogonal basis of the function space, we expand the hyperspectral observation, i.e., $L^x = (L^x_1, \ldots, L^x_p)$ and $\phi(\upsilon)$, in the same functional basis leading to the following:

$$L^x_m(\upsilon) = \sum_{i=1}^{N} c_{mi}\varphi_i(\upsilon), \quad m = 1, p \tag{5}$$

$$\varphi(\upsilon) = \sum_{i=1}^{K} b_i\varphi_i(\upsilon) \tag{6}$$

where $\varphi(\upsilon)$ is the orthogonal basis, N and K are the numbers of this basis, and $\mathbf{c}_m = (c_{m1}, \ldots, c_{mN})'$ and $\mathbf{b} = (b_1, \ldots, b_K)'$ are the coefficient vectors of the basis. Using the basis expansions of (5) and (6), the estimation of $\phi(\upsilon)$ is equivalent to the estimation of $\mathbf{c}_m$ and $\mathbf{b}$. Moreover, a different basis can be selected for L^x and $\phi(\upsilon)$.

Therefore, the basic model of (1) can be reformulated as follows:

$$L^y_m = \int \sum_{i=1}^{K}\sum_{j=1}^{N} b_i c_{mj}\varphi_i(\upsilon)\varphi_j(\upsilon)d\upsilon + \varepsilon. \tag{7}$$

Finally, we can obtain the retrieval result, $\phi(\upsilon)$, by solving the following optimization problem to find the vector coefficient:

$$\hat{\mathbf{b}} = \underset{\mathbf{b}\in\Re^K}{\arg\min} \sum_{m=1}^{p} \left[L^y_m - \int \sum_{i=1}^{K}\sum_{j=1}^{N} b_i c_{mj}\varphi_i(\upsilon)\varphi_j(\upsilon)d\upsilon \right]^2. \tag{8}$$

The selection of the functional basis and its parameter require comprehensive consideration of the fitting capability and computational efficiency based on the spectral distribution characteristics of corresponding variables. In general, the choice of the orthogonal basis is not a key step in the final estimate. Here we used the Fourier basis as follows:

$$\varphi_i(\upsilon) = \begin{cases} 1 & \text{if } i = 1 \\ \sin\left(\dfrac{i}{2}\omega\upsilon\right) & \text{if } i \text{ is even} \\ \cos\left(\dfrac{i-1}{2}\omega\upsilon\right) & \text{otherwise.} \end{cases} \tag{9}$$

The basis number parameters N and K, defined in (5) and (6), are fixed constants. In general, smoother SRF functions should have smaller K values. Normally, the SRF is not a jagged curve, and a small K value is thus appropriate. In this study, N should be larger than K. As L^x is known, larger values of N should theoretically yield a higher fitting accuracy based on (5). However, considering the calculation efficiency and overfitting problem, a value of N under 100 is sufficient. The selection of optimal K and N values will be discussed in Section IV. The coefficient matrix $\mathbf{c}_m = (c_{m1}, \ldots, c_{mN})'$ is computed by the least squares method.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:21:19 UTC from IEEE Xplore. Restrictions apply.

3) Solution: We used matrix notations to simplify the equations. Let vector $C = (\mathbf{c}_1, \ldots, \mathbf{c}_p)$ and $B = (b_1, \ldots, b_K)'$ (equivalent to $\mathbf{b}$) represent the basis coefficients and $\Phi = (\varphi_1(\upsilon), \ldots, \varphi_N(\upsilon))'$ and $\Psi = (\varphi_1(\upsilon), \ldots, \varphi_K(\upsilon))'$ represent the Fourier bases. Therefore, (5)–(7) can be rewritten as follows:

$$L_m^x = C \cdot \Phi \tag{10}$$

$$\varphi = B'\Psi \tag{11}$$

$$L^y = C J_{\Phi\Psi'} B \tag{12}$$

where $J_{\Phi\Psi'} = \int \Phi\Psi' d\upsilon$.

Using (10)–(12), we can solve (8) as a standard multiple regression problem as follows:

$$\hat{B} = \underset{B\in\Re^K}{\arg\min} \left\| L^y - C J_{\Phi\Psi'} B \right\|^2 \tag{13}$$

$$\hat{\varphi}(\upsilon) = \sum_{i=1}^{K} \hat{b}_i \varphi_i(\upsilon). \tag{14}$$

To achieve a more physical retrieval, the solution to (13) should also satisfy several constraints that are as follows.

1) Pointwise summation normalization: $\int \varphi(\upsilon) d\upsilon = 1$.
2) Nonnegativity: $\forall \upsilon, \phi(\upsilon) \geq 0$.
3) Band limitedness: $\varphi(\upsilon) = 0\,, \upsilon \geq \upsilon_{\max}$ or $\upsilon \leq \upsilon_{\min}$.

The second and third constraints are used to avoid secondary peaks and oscillations in the tails of the retrieved SRFs. In this study, $\upsilon_{\max}$ and $\upsilon_{\min}$ are set adaptively instead of via specification by users. For each retrieval, we performed two rounds or calculations. The band limitedness was automatically determined based on the first round of retrieval results, which is the spectral position where the response value is first reduced from the maximum value to below 0.5%. The choice of this threshold is based on the consideration that it can effectively remove the false information from the tail without losing the accuracy of the convolution.

Using these constraints, the optimization of (13) is more complicated and it is difficult to obtain its analytical solution. We note the first constraint can be simplified to $\sum_{i=1}^{s}\sum_{j=1}^{K} b_j \varphi_j(\upsilon_i) = 1$, which represents a hyper plane in K-dimensional Euclidean space. Similarly, we assume that there are enough finite points to stratify the second constraint, such as $\upsilon_1, \ldots, \upsilon_s$, for the actual solution. These constraints are then denoted as

$$\mathbf{B} = \left\{ B : \forall i = 1, \ldots, s, B'\Phi(\upsilon_i) \geq 0, \sum_{i=1}^{s} B'\Phi(\upsilon) = 1 \right\}. \tag{15}$$

The optimization of (13) can then be refined as follows:

$$\hat{B} = \underset{B\in\mathbf{B}}{\arg\min} \left\| L^y - C J_{\Phi\Psi'} B \right\|^2. \tag{16}$$

This is a convex optimization and numerous robust and efficient methods have been developed to approach this issue. In this study, we used the dual proximal gradient method to solve (16) [32]. Ultimately, Table I summarizes the proposed SRF retrieval algorithm as pseudocode. Steps 1 and 2 are data preprocessing, where hyperspectral data is fitted to a set of smooth functions based on Fourier basis. Steps 3 and 4 are the construction of retrieval model, and step 5 is to find the constrained solutions of the retrieval model. According to the first-round solution, steps 6 and 7 automatically determine the band limitation, then repeat step 5 to obtain the final estimation. Note that Table I only shows the processing steps of the retrieval algorithm and does not include the data acquisition process.

TABLE I
PSEUDOCODE FOR THE PROPOSED SRF RETRIEVAL ALGORITHM

Input: Data from experiment $L_m^y, L_m^x(\nu_j), m = 1, \ldots, p, j = 1, \ldots, s$.
Output: Estimation of $\varphi(\nu)$ denoted $\widehat{\varphi}(\nu)$
Step1: Choose the parameter N.
Step2: Smooth the hyperspectral radiance $L_m^x(\nu), m = 1, \ldots, p$ by fourier basis:
for $m = 1, \ldots, p$ **do**

$$\mathbf{c}_m \leftarrow \underset{\mathbf{c}_m\in\mathbb{R}^N}{\arg\min} \left[\sum_{j=1}^{s} \left(L_m^x(\nu_j) - \sum_{i=1}^{N} c_{mi}\phi_i(\nu_j) \right)^2 \right]$$

$$L_m^x(\nu) \leftarrow \textstyle\sum_{i=1}^{N} c_{mi}\phi_i(\nu)$$

end
Step3: Choose the parameter K
Step4: Set $\varphi(\nu) \leftarrow \sum_{i=1}^{K} b_i\phi_i(\nu)$ and the constraint set $\mathbf{B}$ in (15)

$$\text{Step5: } \widehat{B} = \underset{B\in\mathbf{B}}{\arg\min} \left[\sum_{m=1}^{p} \left(L_m^y - \sum_{i=1}^{K} b_i c_{mi} \right)^2 \right]$$

Step6: Determine the band limitation $\nu_{\min}$ and $\nu_{\max}$ based on $\widehat{\varphi}(\nu) = \sum_{i=1}^{K} \widehat{b}_i\phi_i(\nu)$
Step7: $\forall j = 1, \ldots, s$,retain the $\nu_j \in [\nu_{\min}, \nu_{\max}]$ and repeat the Step5 with these spectral position inside the band limitation to get the estimation of φ.
Return $\widehat{\varphi}(\nu) = \sum_{i=1}^{K} \widehat{b}_i\phi_i(\nu)$

III. EXPERIMENTAL VALIDATION AND SOME RESULTS

To verify the effectiveness of the proposed method, we conducted both simulated and real data experiments. Simulated data experiments were performed to validate the effectiveness of the proposed algorithm for different SRF situations. Real data experiments were performed based on real observations to explore the applicability of the method to real circumstances.

A. Simulated Data Experiments

Assuming ideal conditions, (1) is strictly true. The multispectral observation, L^y, was replaced with the simulated radiance by convolving the METOP-B IASI spectra with the prelaunch SRF of a given band. Here the ε denoted as $\varepsilon_{y,\text{rand}}$ only contains the multispectral instrument's noise which is a random vector with an average of zero and standard deviation of the instrument's nominal NEdR. In the simulation experiments, the prelaunch SRF was regarded as the true value for the retrieval solution. In this study, a daily global observation of the METOP-B IASI was used for simulation, which was downsampled to reduce the redundancy of adjacent scenes. WV absorption bands centered at 6.5 and 7.2 μm, window

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:21:19 UTC from IEEE Xplore. Restrictions apply.

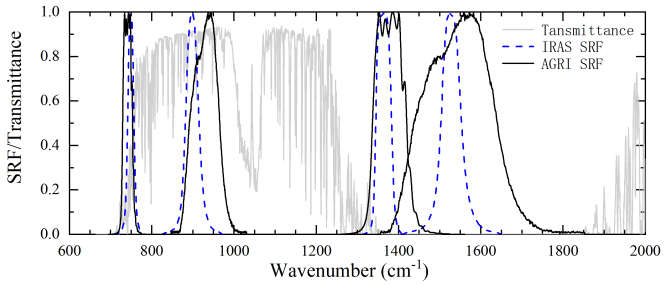


Fig. 2. Spectral response distributions of FY-3C IRAS (black dashed lines) and FY-4A AGRI (gray solid lines) selected bands for simulated data experiments as well as the simulated TOA spectra of the transmittance (light gray solid line) over a clear sky.

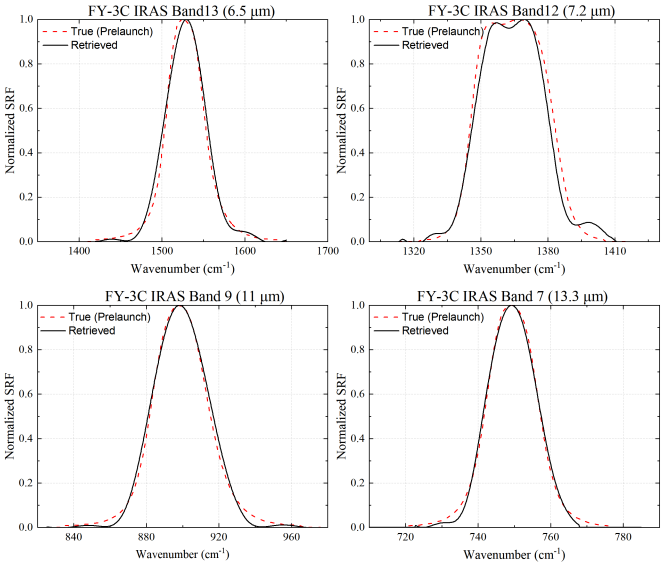


Fig. 3. Retrieved SRFs for FY-3C IRAS compared with the prelaunch measurements based on simulated data with random noise.

bands centered at 11 μm, and CO_2 absorption bands centered at 13.3 μm in the FY-3C IRAS and FY-4A AGRI were selected for retrieval. IRAS is a narrowband sounder mounted on a polar-orbit platform [33] while AGRI is a broadband imager mounted on a geostationary platform [34]. Fig. 2 shows the prelaunch SRF distributions of these selected bands as well as the simulated TOA spectra of the transmittance over a clear sky. The SRFs of IRAS are denoted by black dashed lines, whereas those of AGRI are denoted by gray solid lines. These eight SRFs have different shapes, widths, and center positions, which allow us to investigate the effectiveness of proposed method for different SRF distributions.

Based on the previously mentioned simulation, Figs. 3 and 4 show the SRF retrieval results for IRAS and AGRI compared with the prelaunch measurements, respectively. We can observe that, regardless of the SRF bandwidth and absorption intensity of the corresponding band (as long as it is a unimodal distribution), the proposed method yields ideal retrievals. The retrieved SRFs for IRAS band 12 and AGRI bands 9 and 10 slightly deviate from the true value, which are characterized by a bimodal distribution. For these bands, Fig. 5 illustrates the global brightness temperature differences between the multispectral convolutions of the IASI spectra with the retrieved

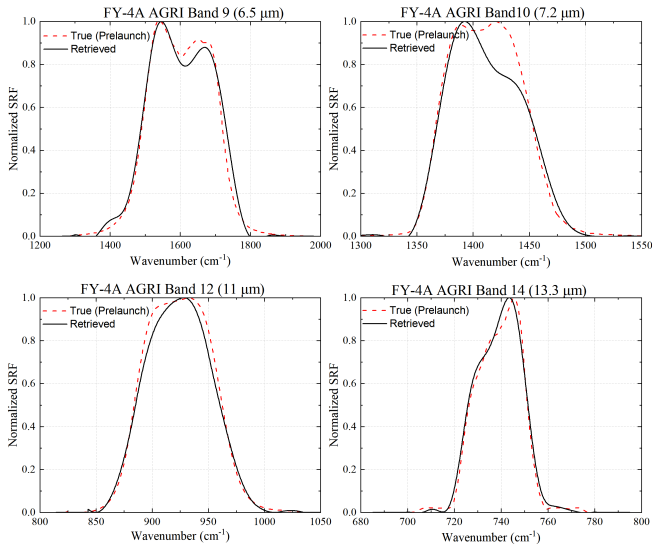


Fig. 4. Retrieved SRFs for FY-4A AGRI compared with the prelaunch measurements based on simulated data with random noise.

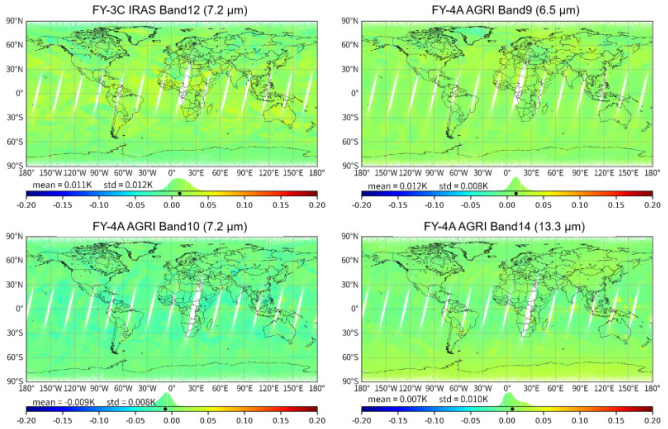


Fig. 5. Global brightness temperature differences between the multispectral convolutions of IASI spectra with the retrieved SRF and the prelaunch measurement.

SRF and the prelaunch measurement. The differences between the retrieved and the true SRFs do not induce substantial scene-dependent biases, where the global bias distribution of these three bands is comparable with that of band 14 in the FY-4. The mean biases caused by retrieval error are less than 0.012 k, and standard deviations of the biases are also less than 0.012 k, which are small enough compared with the instrument's nominal noise specification of 0.12–0.4 K. The result here does not indicate that the shape is not important. In fact, through retrieval, we obtained an optimal SRF shape which is almost equivalent to the truth. Although the shape is not exactly the same as the true value (which is not known in many cases), it can achieve good application effects, which may, in many cases, hardly be achieved by shifting correction.

To explore why we were unable to achieve an ideal retrieval for the bimodal situation, Fig. 6, using AGRI as an example, presents another simulation experiment result in which the simulated observation is noise free, i.e., $\varepsilon_{y,\mathrm{rand}} = 0$. In this instance, K has a large value of 50. Fig. 6 indicates that, when there is no noise, the algorithm can obtain an exact

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:21:19 UTC from IEEE Xplore. Restrictions apply.

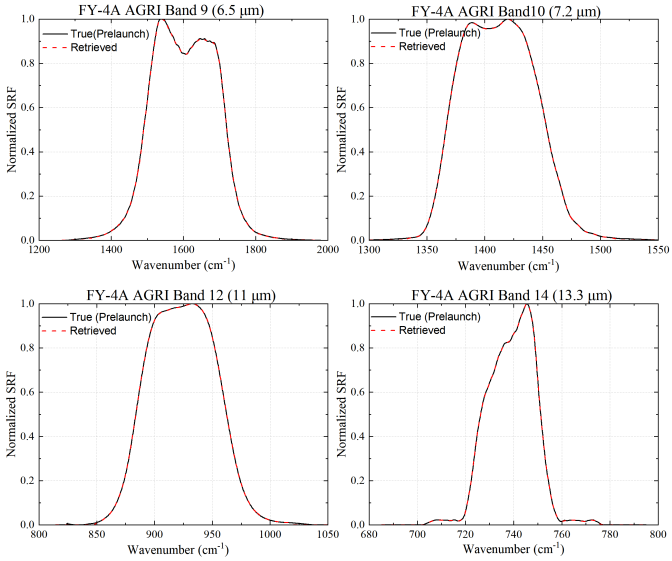


Fig. 6. Retrieved SRFs for FY-4A AGRI compared with the prelaunch measurements based on simulated data without noise.

retrieval of the AGRI's SRFs, which suggests that the proposed FDA retrieval is mathematically feasible. According to the basic principles of FDA, for the function of the bimodal or multimodal distribution, the value of K must theoretically be larger than that of the unimodal to obtain a better fitting effect. For noise-free data, K can be infinite. However, when there is noise in the data, K can only be a finite value, where an excessive value renders the influence of the noise prominent which leads to overfitting and distortion. This is why the bimodal distribution SRF cannot yield perfect results. Using a limited value of K can suppress the influence of the noise and ensure a stable solution, which is also the advantage of our method. Nevertheless, the perfect solution cannot be obtained; however, we can obtain an optimal equivalent solution.

B. Real Data Experiments

1) Data Collection: We have two choices to obtain the matched spectra radiance. One is based on a radiative transmission simulation and the other is based on intercalibration collocation. The simultaneous nadir overpass (SNO) pairs of the multispectral radiometer and hyperspectral sounder over the same targets were collected for retrieval. The SNO avoids errors induced by the radiative transmission model and its input parameters. The collocated pixels should be spatially collocated, temporally concurrent, and geometrically aligned. The collocation technique follows the recommendations of global space-based intercalibration system (GSICS) [35]–[38]. The spatial distance between the centers of the pixel pairs was set to less than half of the instantaneous field of view (IFOV) of the multispectral imager. The time difference was limited to less than 5 min. Both satellite viewing zenith angles (VZAs) were limited to 30° to maintain a near nadir observation, whereas the relative difference of the cosine VZA was restricted to less than 1%. In addition, to further reduce the collocation biases caused by mismatching, geolocation error, and significant differences in spatial resolution, a spatially uniform filter was applied according to the coefficient of

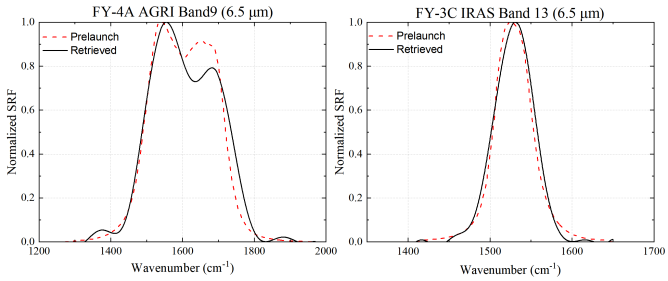


Fig. 7. Retrieved SRFs for (Left) FY-4A AGRI and (Right) FY-3C IRASI 6.5-μm bands.

variation (CV) of the multispectral observations within the IASI IFOV and its corresponding environment region (three times that of the IASI IFOV). The CV should be less than 1%.

In our study, the observations from FY-3C IRAS and FY-4A AGRI were collocated with METOP-B IASI. For the geostationary FY-4A satellite, the SNO collocated pixels were distributed within latitudes of ±30°. For the polar orbiting FY-3C satellite, the SNO observations generally occurred in the high-latitude polar region where the surface and atmosphere are homogeneous. To ensure a representative result, we extended the high-latitude SNOs to the low-latitude tropical region using near-coincidence paralleled orbits with a nadir distance over the equator within 30 km, referred to as SNOx [39], [40]. As a result, we can obtain global collocations involving a wider variety of land-cover types and higher absorptive gas concentrations.

2) Retrievals: Real retrievals based on the collocated observations were conducted for the IRAS and AGRI 6.5-μm WV bands, which both have relatively minor spectral and radiometric calibration errors. This study does not involve the separation of radiometric and spectral calibration errors, which is important in practical applications. Therefore, we chose the band with smaller calibration error for the practical experiment to avoid a further interrogation of this issue. This band was also convenient for the feasibility exploration of the retrieval algorithm, because in this case the retrieved SRF was expected to be similar to the prelaunch measurement.

Fig. 7 shows the SRF retrieval results based on the SNO and SNOx collocations, whereas Fig. 8 compares the scene-dependent radiance and brightness temperature (TBB) biases of these two bands assessed using the prelaunch and retrieved SRF. The collocated pixels of AGRI are from January to July 2019 whereas those of IRAS are from March to July 2019, where there were more SNOx orbits between FY-3C and METOP-B. Both retrievals used more than 5.000 collocations.

The retrieved SRF of IRAS band 13 is highly similar to the prelaunch measurement, and has a slight correction to original systematic and scene-dependent biases. For AGRI band 9, the intercalibration assessments in Fig. 8 show that the operational calibration have a scene-dependent bias, with a systematic bias of approximately 0.2 K. A clear modification appears between the prelaunch measurement and retrieved result. This modification may be partly attributed to the limitations of the algorithm itself for the bimodal problem, and may partly be a correction effect to the prelaunch error. As illustrated in Fig. 8, the retrieved SRF can substantially reduce the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:21:19 UTC from IEEE Xplore. Restrictions apply.

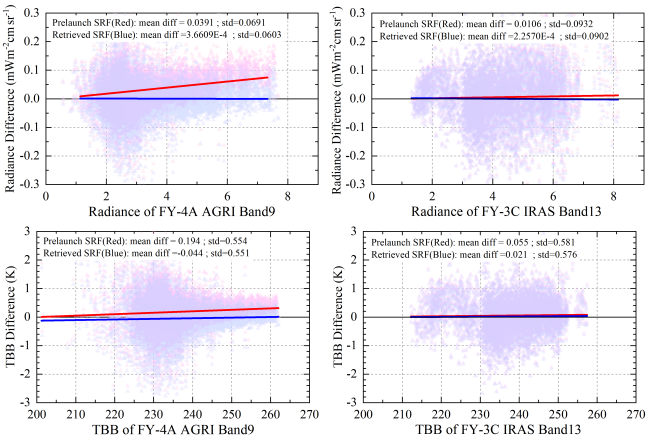


Fig. 8. Comparison of the scene-dependent radiance and brightness temperature (TBB) biases for (Left) FY-4A AGRI and (Right) FY-3C IRAS 6.5-μm bands based on the prelaunch and retrieved SRF using intercalibration collocations.

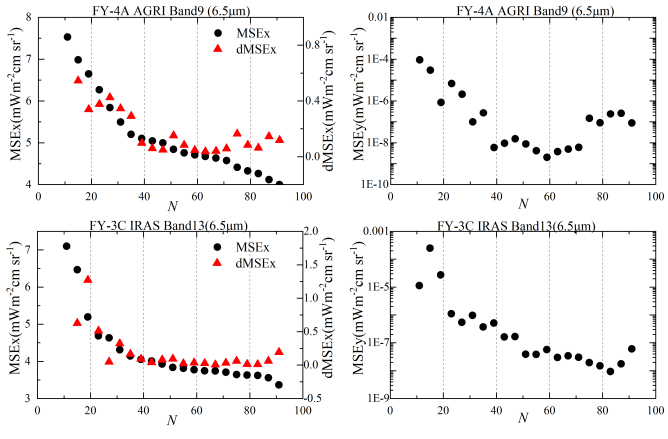


Fig. 9. Variation in the mean square error of (Left) functional approximated radiance spectra (MSEx) and its first difference (dMSEx) versus N as well as (Right) corresponding mean square error (MSEy) of the convoluted radiance of $\hat{L}^x$ with the prelaunch SRF.

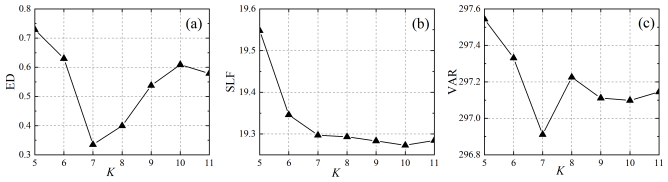


Fig. 10. Variations in the retrieval performance of the FY-3C IRAS band 13 versus k. (a) ED factor of the retrieved SRF. (b) SLF of the retrieval model. (c) Dispersion index (VAR) of the predicted multispectral radiance.

systematic and scene-dependent biases, thereby decreasing its dispersion (the standard deviations) and rendering the radiation more consistent. Therefore, the retrieval results should be more reasonable than the prelaunch measurements.

IV. Discussion

N and K are two important parameters in the proposed FDA retrieval model, that are used to functionally characterize the hyperspectral observations and SRF. Different values of N and K likely yield different solutions. To explore how parameter selection affects the retrieval, sensitivity assessments were performed based on the simulated data with noise. According to the sensitivity behaviors, we also discuss the approaches to determine the optimal N and K values using quantitative factors.

A. N Selection

The functional approximation smooths the discrete observations. Taking the high absorptive WV bands of AGRI and IRAS as an example, Fig. 9 illustrates the bias variation in the functional approximated radiance spectra (denoted as $\hat{L}^x$) versus N as well as its influence on the convolution radiance. In this instance, N takes every four values, ranging from 11 to 91. MSEx is the mean square error of $\hat{L}^x$ compared with L^x, calculated as follows:

$$\mathrm{MSE}_x = \frac{1}{s * p} \sum_{i=1}^{s} \sum_{j=1}^{p} \left(\hat{L}_{ij}^x - L_{ij}^x\right)^2. \tag{17}$$

dMSEx is the first difference of MSEx and MSEy is the mean square error of the convolution radiation of $\hat{L}^x$ with the prelaunch SRF. dMSEx is a useful indicator that can characterize the effect of overfitting to a certain extent. In Fig. 9(a), the MSEx of AGRI band 9 strictly decreases with an increasing N value, but the slope changes at $N = 39$. At this point, dMSEx is mostly constant and near zero, whereas when $N > 71$, dMSEx begins to increase again. Fig. 9(b) shows that, when N ranges from 39 to 71, the convolution errors are smaller; however, when $N > 71$, MSEx decreases and MSEy increases which leads to an overfitting approximation. Therefore, for this band, an optimal N value can be between 39 and 71. The behavior of IRAS band 13 presents similar characteristics. To a certain extent, the retrieval is not sensitive to the value of N, as the effect of different N values on the convolution error is insignificant. Therefore, we can empirically set N to a given constant, such as 65.

B. K Selection

The wider the target SRF's bandwidth and the more the fluctuation in its shape, the more sensitive the retrieval is to the K value, and the retrieval becomes increasingly difficult. For different SRF cases, we should select different optimal K values. Taking IRAS band 13 as an example, Fig. 10 illustrates the dependence of the retrieval performance from K. Owing to the availability of true data in the simulated experiments, three quantitative factors were used to provide an overall quantitative assessment, i.e., the ED factor, square loss function (SLF), and dispersion index (VAR). The ED characterizes the consistency between the retrieved SRF and true value, and the SLF represents the accuracy of the retrieval model; these are respectively defined as follows:

$$\mathrm{ED} = \|\phi_{\mathrm{retrieved}} - \phi_{\mathrm{true}}\| \tag{18}$$

$$\mathrm{SLF} = \left\|L^y - \hat{L}^y\right\|^2 \tag{19}$$

where $\phi_{\mathrm{retrieved}}$ and ϕ_{true} are the retrieved and true SRF, respectively, and $\hat{L}^y$ is the predicted multispectral radiance simulated by convoluting L^x with $\phi_{\mathrm{retrieved}}$. VAR is the variance of $\hat{L}^y$ reflecting the dispersion of the predicted results. Smaller factors yield better retrieval performance.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:21:19 UTC from IEEE Xplore. Restrictions apply.

In Fig. 10(a), we observe for this case that the best retrieval result can be obtained at $K = 7$, where the retrieved SRF is closest to the true value. Fig. 10(b) and (c) shows that when $K \leq 7$ and increases, both SLF and VAR decrease. If we continue to increase the K value, SLF continues to decrease as its variation becomes constant and there is an increase in the dispersion of the predicted multispectral radiance, which is a typical feature of overfitting. Fig. 10 provides a method to quantitatively determine the optimal K value, which should be a value that produces a balance between the accuracy of the retrieval model and dispersion of the predicted results. The K value can be determined based on the SLR and VAR as follows:

$$K = \arg\min_{5 \leq K \leq 11} \{\nabla \mathrm{SLF} + \mathrm{VAR}\}. \tag{20}$$

V. Conclusion

We proposed an innovative new in-flight SRF retrieval method for the multispectral radiometer based on intercomparisons via FDA technology. By considering all the discrete observations as smooth functions, the hyperspectral radiance and SRF were expanded using the Fourier functional basis. The convolution formula was converted from a normally pointwise summation to a functional integration, which ensured that the variable was transformed from numerous SRF samples to several function parameters, thereby successfully avoiding the ill-posed problem. The algorithm has no penalty constraints and does not require initial values for optimal solutions.

Based on both the simulated and real data experiments, the proposed algorithm was applied to multiple thermal infrared bands of FY-3 IRAS and FY-4 AGRI to confirm its effectiveness and applicability. In the simulation experiments, we verified the algorithm's reliability and its ability to obtain exact retrievals when the data are noise-free. When the data contains noise, despite an improved algorithm performance for the unimodal distribution SRF as compared with the bimodal distribution, an acceptable equivalent solution can be obtained with a negligible error in convolution radiance. Using real observations from the SNO and SNO_X intercalibration collocations with METOP-B IASI, the results based on the real data experiments indicate that the retrieved SRF is similar but more reasonable when compared with the reference, which can substantially reduce existing scene-dependent biases and renders the multispectral radiation more consistent with that of the IASI.

Currently, this study mainly focused on exploring the feasibility of the retrieval method itself. However, if we intend to apply this algorithm to practical situations, we must still consider several specific issues. First, the radiometric and spectral calibration biases should be separated in advance. Otherwise, the optimal solution of our algorithm will attribute all biases to spectral error, and the result may not be credible. Second, the effect that changes in the SRF have on the radiometric calibration was not investigated in this study. If the SRF changes, the radiometric calibration result will be adjusted accordingly. Therefore, we must examine the mutual feedback effect of the two sides. Uncertainty is also an important issue. Both the retrieval methodology and the observation data noise introduce uncertainty. Based on the uncertainty theory of the least squares method, we can derive the theoretical uncertainty of the retrieval methodology. However, because our problem is more complicated than common least squares regression situations, the derivation result is likely to be unstable. For the observation data noise, the proposed method has substantial advantages in restraining the uncertainty introduced by random noise, but systematic errors and spectra-correlated errors also produce uncertainty. These issues are complex and subject to further research. Future studies will focus on the further substantiation of our algorithm by combining it with practical requirements and its application to more cases.

In summary, this method is a practical application of GSICS intercalibration that is suitable for postlaunch spectral correction and has the additional potential for operational spectral calibration and support for climate studies. Although the results in this study are particularly relevant to infrared spectra, the algorithm may be also applicable to other spectral bands in future.

Acknowledgment

The authors would like to thank two anonymous reviewers for their critical review and judicious comments that promoted the quality of this article. They also thank the members of the GSICS IR Subgroup for useful conversation and intercalibration technology supporting. They are particularly grateful to Dr. Jun Li of the University of Wisconsin–Madison for the valuable suggestions and discussions on the retrieval model.

References

[1] B. J. Sohn *et al.*, "Possible shift of spectral response function of the MODIS 6.8 μm water vapor channel causing a cold bias up to 3K," *Atmos. Meas. Tech. Discuss.*, vol. 3, no. 4, pp. 3277–3291, 2010, doi: 10.5194/amtd-3-3277-2010.

[2] B. Sohn, H. Park, H. Han, and M. Ahn, "Evaluating the calibration of MTSAT-1R infrared channels using collocated Terra MODIS measurements," *Int. J. Remote Sens.*, vol. 29, no. 10, pp. 3033–3042, May 2008, doi: 10.1080/01431160701408394.

[3] D. C. Tobin, H. E. Revercomb, C. C. Moeller, and T. S. Pagano, "Use of atmospheric infrared sounder high–spectral resolution spectra to assess the calibration of moderate resolution imaging spectroradiometer on EOS aqua," *J. Geophys. Res.*, vol. 111, no. 9, pp. 1–15, 2006, doi: 10.1029/2005JD006095.

[4] C. Cao, M. Goldberg, and L. Wang, "Using IASI observations to resolve HIRS spectral response function induced intersatellite biases," *Proc. SPIE*, vol. 7085, Sep. 2008, Art. no. 708505.

[5] C. Cao, M. Goldberg, and L. Wang, "Spectral bias estimation of historical HIRS using IASI observations for improved fundamental climate data records," *J. Atmos. Ocean. Technol.*, vol. 26, no. 7, pp. 1378–1387, Jul. 2009, doi: 10.1175/2009JTECHA1235.1.

[6] D. W. Hillger and T. J. Schmit, "Observing systems: The GOES-13 science test: A synopsis," *Bull. Amer. Meteorol. Soc.*, vol. 90, no. 5, pp. 592–597, May 2009, doi: 10.1175/2008BAMS2564.1.

[7] T. J. Schmit and M. Gunshor, "Apparent cold bias in the GOES-13 imager band 6 (13.3 μm)," Technical Memorandum, Tech. Rep., 2007.

[8] D. Hillger *et al.*, "The GOES-15 science test: Imager and Sounder radiance and product validations," NOAA, Tech. Rep., vol. 141, 2011, Art. no. GOES515.

[9] X. Wu and F. Yu, "Correction for GOES imager spectral response function using GSICS. Part I: Theory," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 3, pp. 1215–1223, Mar. 2013, doi: 10.1109/TGRS.2012.2236100.

[10] F. Yu and X. Wu, "Correction for GOES imager spectral response function using GSICS. Part II: Applications," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 3, pp. 1200–1214, Mar. 2013.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:21:19 UTC from IEEE Xplore. Restrictions apply.

[11] T. J. Hewison and J. Muller, "Ice contamination of Meteosat/SEVIRI implied by intercalibration against Metop/IASI," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 3, pp. 1182–1186, Mar. 2013, doi: 10.1109/TGRS.2012.2236335.
[12] L. Wang, X. Wu, F. Weng, and M. D. Goldberg, "Effects of ice decontamination on GOES-12 imager calibration," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 3, pp. 1224–1230, Mar. 2013, doi: 10.1109/TGRS.2012.2225839.
[13] F.-M. Bréon, D. Jackson, and J. Bates, "Evidence of atmospheric contamination on the measurement of the spectral response of the GMS-5 Water vapor channel," *J. Atmos. Ocean. Technol.*, vol. 16, no. 11, pp. 1851–1853, Nov. 1999.
[14] R. P. Galvin, "Imager SN08 (GOES-13) spectral shift inquiry (GIR NOPS-0004)," ITT Ind. Tech. Memorandum NOPQ DCN 698, vol. 3, 2007, pp. 20–25.
[15] X. Na, H. Xiu-Qing, and C. Lin, "Evaluation and correction of atmosphere contamination on the measurement of the spectral response of FY-2 water vapor channels," *Spectrosc. Spectral Anal.*, vol. 37, no. 8, pp. 2325–2329, 2017.
[16] R. Chen, C. Cao, and W. P. Menzel, "Interstellite calibration of NOAA HIRS CO_2 channels for climate studies," *J. Geophys. Res., Atmos.*, vol. 118, no. 11, pp. 5190–5203, Jun. 2013.
[17] H. Sun, W. Wolf, C. Barnet, L. Zhou, and M. Goldberg, "MODIS infrared channel spectral response function calibration with co-located AIRS observation," in *Proc. Symp. Meteorol. Observ. Instrum.*, 2007, pp. 1–6.
[18] R. Quast, R. Giering, Y. Govaerts, F. Rüthrich, and R. Roebeling, "Climate data records from meteosat first generation part II: Retrieval of the in-flight visible spectral response," *Remote Sens.*, vol. 11, no. 5, p. 480, Feb. 2019, doi: 10.3390/rs11050480.
[19] J. O. Ramsay and B. W. Silverman, *Applied Functional Data Analysis: Methods and Case Studies*, 2nd ed. Springer, 2002.
[20] J. O. Ramsay and B. W. Silverman, *Functional Data Analysis*, 2nd ed. Springer, 2005.
[21] J. Ramsay, *Functional Data Analysis With R and MATLAB*. Springer, 2009.
[22] J. L. Wang, J. M. Chiou, and H. G. Müller, "Review of functional data analysis," *Statistics*, 2015, doi: 10.1080/02331888.2014.886689.
[23] C. Liu, S. Ray, G. Hooker, and M. Friedl, "Functional factor analysis for periodic remote sensing data," *Ann. Appl. Statist.*, vol. 6, no. 2, pp. 601–624, Jun. 2012, doi: 10.1214/11-AOAS518.
[24] P. Das, A. Lahiri, and S. Das, "Understanding sea ice melting via functional data analysis," *Current Sci.*, vol. 115, no. 5, p. 920, Sep. 2018, doi: 10.18520/cs/v115/i5/920-929.
[25] L. Yinshuang and L. I. U. Liming, "PM2.5 pollution characteristic in Beijing-Tianjin-Hebei region based on the perspective of functional data analysis," *Oper. Res. Trans.*, vol. 22, no. 2, pp. 105–114, 2018.
[26] W. Wu *et al.*, "An accurate method for correcting spectral convolution errors in intercalibration of broadband and hyperspectral sensors," *J. Geophys. Res., Atmos.*, vol. 123, no. 17, pp. 9238–9255, Sep. 2018, doi: 10.1029/2018JD028585.
[27] T. J. Hewison, "An evaluation of the uncertainty of the GSICS SEVIRI-IASI intercalibration products," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 3, pp. 1171–1181, Mar. 2013.
[28] J. Li, W. P. Menzel, and A. J. Schreiner, "Variational retrieval of cloud parameters from GOES sounder longwave cloudy radiance measurements," *J. Appl. Meteorol.*, vol. 40, no. 3, pp. 312–330, 2001.
[29] H. Cardot, F. Ferraty, and P. Sarda, "Functional linear model," *Stat. Probab. Lett.*, vol. 45, no. 1, pp. 11–22, 1999, doi: 10.1016/S0167-7152(99)00036-X.
[30] H. Cardot, F. Ferraty, and P. Sarda, "Spline estimators for the functional linear model," *Statistica Sinica*, vol. 13, pp. 571–592, 2003.
[31] P. Hall and J. L. Horowitz, "Methodology and convergence rates for functional linear regression," *Ann. Statist.*, vol. 35, no. 1, pp. 70–91, Feb. 2007, doi: 10.1214/009053606000000957.
[32] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge, U.K.: Cambridge Univ. Press, 2004.
[33] C. Dong *et al.*, "An overview of a new chinese weather satellite FY-3A," *Bull. Amer. Meteorol. Soc.*, vol. 90, no. 10, pp. 1531–1544, Oct. 2009, doi: 10.1175/2009BAMS2798.1.
[34] J. Yang, Z. Zhang, C. Wei, F. Lu, and Q. Guo, "Introducing the new generation of Chinese geostationary weather satellites, Fengyun-4," *Bull. Amer. Meteorol. Soc.*, vol. 98, no. 8, pp. 1637–1658, 2016, doi: 10.1175/BAMS-D-16-0065.1.
[35] X. Wu, T. Hewison, and Y. Tahara, "GSICS GEO-LEO intercalibration: Baseline algorithm and early results," *Proc. SPIE*, vol. 7456, Aug. 2009, Art. no. 745604.
[36] T. J. Hewison *et al.*, "GSICS inter-calibration of infrared channels of geostationary imagers using Metop/IASI," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 3, pp. 1160–1170, Mar. 2013, doi: 10.1109/TGRS.2013.2238544.
[37] N. Xu, L. Chen, X. Hu, L. Zhang, and P. Zhang, "Assessment and correction of on-orbit radiometric calibration for FY-3 VIRR thermal infrared channels," *Remote Sens.*, vol. 6, no. 4, pp. 2884–2897, Mar. 2014, doi: 10.3390/rs6042884.
[38] N. Xu, X. Hu, and M. Min, "Inter-calibration of infrared channels of FY-2/VISSR using high-spectral resolution sensors IASI and AIRS," *J. Remote Sens.*, vol. 16, no. 5, pp. 939–952, 2012.
[39] C. Cao, S. Uprety, and S. Blonski, "Establishing radiometric consistency among VIIRS, MODIS, and AVHRR using SNO and SNOx methods," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2012, pp. 6928–6931.
[40] S. Uprety, "Evaluating radiometric consistency between Suomi NPP VIIRS and NOAA-19 AVHRR using extended simultaneous nadir overpass in the low latitudes," *Proc. SPIE*, vol. 8866, Sep. 2013, Art. no. 88660L.



Na Xu received the B.S. degree in atmospheric science from the Ocean University of China, Qingdao, China, in 2005, and the Ph.D. degree in atmospheric physics and environment from the Institute of Atmospheric Physics, Chinese Academy of Sciences, Beijing, China, in 2010.

She is a Professor with the National Satellite and Meteorological Center, China Meteorological Administration, Beijing, and a member of the Global Space-based Inter-Calibration System (GSICS/WMO). She has been an Instrument Scientist of the Medium Resolution Spectral Imager onboard the Chinese FengYun-3D (FY-3D) and in charge of the calibration and validation for MERSI II, and is the Chief Director of the product generation system (PGS) for FengYun-3 and FengYun-4 Projects. Her research interests include calibration and validation for optical sensors and remote sensing of cloud microphysics.



Peng Wu is pursuing the Ph.D. degree with the Beijing International Center for Mathematical Research, Peking University, Beijing, China.

His main research interests include causal inference, nonparametric/semiparametric models, data fusion, and recommender systems.



Gang Ma is a Research Scientist with the National Satellite Meteorological Center (NSMC), Beijing, China. He was a Deputy Director of Products Generating System (PGS) of the FengYun-3 (FY3) Project from 2010 to 2019. He is a Chief Scientist to develop a fast radiative transfer model to FY2, FY3, and FY4 infrared and microwave sensors. He is also a key member of a team to introduce FY observations into data assimilation of operational Global/Regional Assimilating and Predicting System (GRAPES) of China. He is working on developing a fast radiative transfer model to FY Meteorological Satellite of China and a supporting technology to assimilating observations of FY satellites into NWP model.



Qirui Hu received the bachelor's degree from the School of Statistics, Beijing normal University, Beijing, China. He is pursuing the Ph.D. degree with the Center for Statistical Science and the Department of Industrial Engineering, Tsinghua University, Beijing.

His main research interests include inference for time series, functional data, and nonparametric statistics as well as applications of statistics in remote sensing, environment, and meteorology.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:21:19 UTC from IEEE Xplore. Restrictions apply.



Xiuqing Hu received the B.S. degree in atmospheric science from Nanjing University, Nanjing, China, in 1996, the M.S. degree in cartography and geographical information systems from Beijing Normal University, Beijing, China, in 2004, and the Ph.D. degree in quantitative remote sensing science from the Institute of Remote Sensing Application, Chinese Academy of Sciences, Beijing, in 2012.

He is the Chief Scientist of the ground segment system for FengYun-3 Meteorological Series Satellite and the Team Leader of satellite calibration/validation of the National Satellite Meteorological Center, China Meteorological Administration (CMA), Beijing. Since 2010, he has been a Professor of engineering; and since 2015, he has been among the selected Technological Leading Talent of CMA. His research interests include calibration and validation of optical and infrared sensors, the retrieval algorithm of aerosol/dust and water vapor, and the fundamental climate data records from environment satellite.



Ronghua Wu is a Senior Engineer of the National Satellite Meteorological Center, China Meteorological Administration (NSMC/CMA), Beijing, China. He is also working with FY-3 data preprocessing, including FY-3B geo-location and FY-3C/3D VIS-NIR radiation calibration. His research interest includes the research and application of lunar calibration and cross calibration.



Yunfeng Wang received the Ph.D. degree in meteorology from the Department of Atmospheric Sciences, Nanjing University, Nanjing, China, in 1999.

From 2002 to 2004, he worked as a Postdoctoral Fellow with the Institute of Atmospheric Physics, Chinese Academy of Sciences, Beijing, China. He has been doing satellite data assimilation and numerical simulation experiments for a long time.



Hanlie Xu received the B.S. degree in atmospheric science from the Nanjing University of Information Science and Technology, Nanjing, Jiangsu, China, in 2008, and the Ph.D. degree in meteorology from the Institute of Atmospheric Physics, Chinese Academy of Sciences, Beijing, China, in 2013.

Her research interests mainly aimed at radiometric calibration and validation for infrared sensors.



Lin Chen received the Ph.D. degree in atmospheric physics and environment from the Chinese Academy of Sciences, Beijing, China, in 2009.

He works as an Associate Professor with the National Satellite Meteorological Center, China Meteorological Administration, Beijing. He is the Deputy Chief Scientist of the ground segment system for FengYun-3 Meteorological Satellite and is a member of the Global Space-based Inter-Calibration System (GSICS/WMO).



Peng Zhang received the M.S. degree in atmospheric optics from the Anhui Institute of Optics and Fine Mechanics, Chinese Academy of Sciences (CAS), Hefei, China, in 1995, and the Ph.D. degree in atmospheric physics from the Institute of Atmospheric Physics, CAS, Beijing, China, in 1998.

From 1998 to 2001, he was with the Earth Observation Research Center, National Space Development Agency, Tokyo, Japan, with a postdoctoral position for the Global Imager/Advanced Earth Observing Satellite (GLI/ADEOS) Project. Since 2001, he has been with the National Satellite Meteorological Center, China Meteorological Administration, Beijing, where he was an Associate Professor from November 2001 to November 2005, has been a Professor since December 2005, and has been one of the leading project scientists of the Chinese meteorological polar-orbit-satellite FengYun 3 (FY-3) series since 2007. He is intensively involved in conceiving, developing, and operating the FY-3-satellite ground segment. His major field focuses on atmospheric remote sensing and satellite data assimilation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:21:19 UTC from IEEE Xplore. Restrictions apply.